



Backup Verification Simulator

Automating Data Integrity & Reliability

 Python Platform

 Automated Validation

 Enterprise Dashboard

• Project Overview

Primary Purpose

- **Verify Integrity:** Fully automated backup verification check.
- **Detect Corruptions:** Find corrupted files before restore.
- **Continuous Monitoring:** Real-time status dashboards.

Integrated Solution

- **Comparison Engine:** Highly efficient byte comparison.
- **Alert System:** Instant alert management module.

Verify backups before disaster recovery.

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SOLID ICON

DATA CENTER

Modern minimal style and perfect pixel icons



DATA CENTER



SECURITY



HOSTING



ROUTER



MAINTENANCE



BIG DATA



DATA TRANSFER



DATABASE



CLOUD STORAGE



DATA PROCESSING



BACKUP



INFRASTRUCTURE



NETWORK



CLOUD COMPUTING



COLOCATION



HOSTING



VIRTUALIZATION



BANDWIDTH



SERVER



SERVER CHECK

• Existing Challenges

⚠ System Vulnerabilities

- **Unverified Backups:** Blind trust in successful logs.
- **File Corruptions:** Silent write failures inside repositories.
- **Manual Validation:** Labor-intensive and highly error-prone.
- **No Visual Tracking:** Lack of real-time health indicator.

Key Operational Risks

- Recovery failure, unexpected data loss, and costly downtime.

Backups are useless if they cannot be restored.



DATA UNAVAILABLE

• Project Objectives



Automate & Detect

Eliminate manual operational errors while immediately flagging repository mismatches.



Validate & Report

Run deeply comprehensive validation checks and output clear reports to operators.



Monitor & Optimize

Maximize infrastructure visibility and significantly reduce daily IT administrative tasks.

Automation improves backup reliability.

• Core Features

Dashboard Monitoring

Continuous dynamic status mapping, visual tracking widgets, and overall simulator workflow state display.

Integrity Verification

Extremely granular verification engine. High-speed file-matching algorithm to confirm bit-level precision.

Advanced Analytics

Complete historical success metrics logging, validation failure records, and file health ratios analysis.

Failure Alerts & UI

Instant visual alert flashing when failure occurs. Modern enterprise-ready application layout interface.

- ## Technology Stack

Application Backend

Python Engine — `os`, `datetime` standard libraries

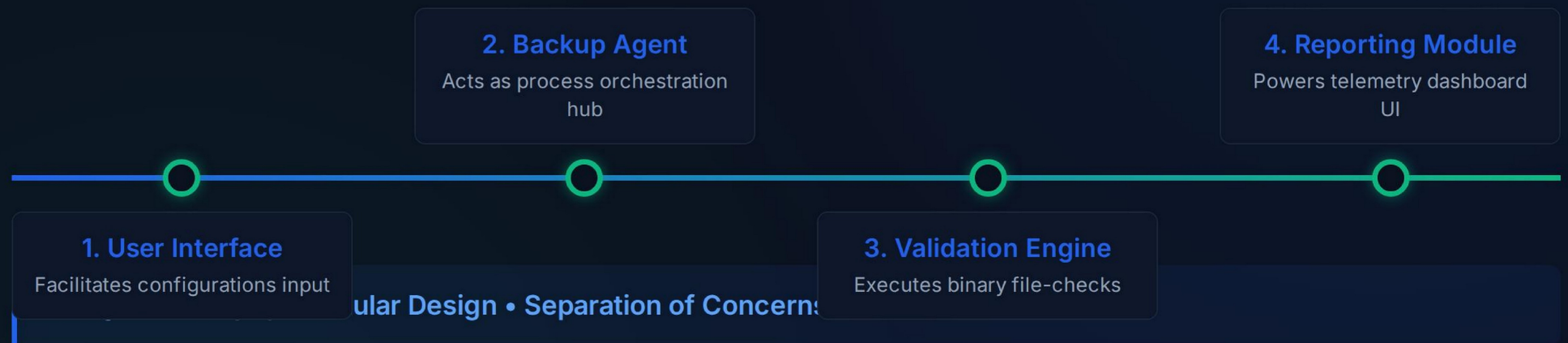
Graphical Front-end

CustomTkinter Framework — High fidelity dark UI templates

Collaborative Tools

VS Code IDE, Git versioning, GitHub workspace

• Architecture Overview



- **Core Modules**



main.py

The bootstrap sequence block.
Directs primary setup and
configures runtime pipelines.



ui.py

Holds layout definitions. Handles
CustomTkinter event bindings and
live updates.



backup.py

The safe data handler. Manages
local folder definitions and reads
source files safely.

- ## Verification Modules



validator.py

The algorithmic brain. Operates fast byte-matching checking against checksum records.



backup_agent.py

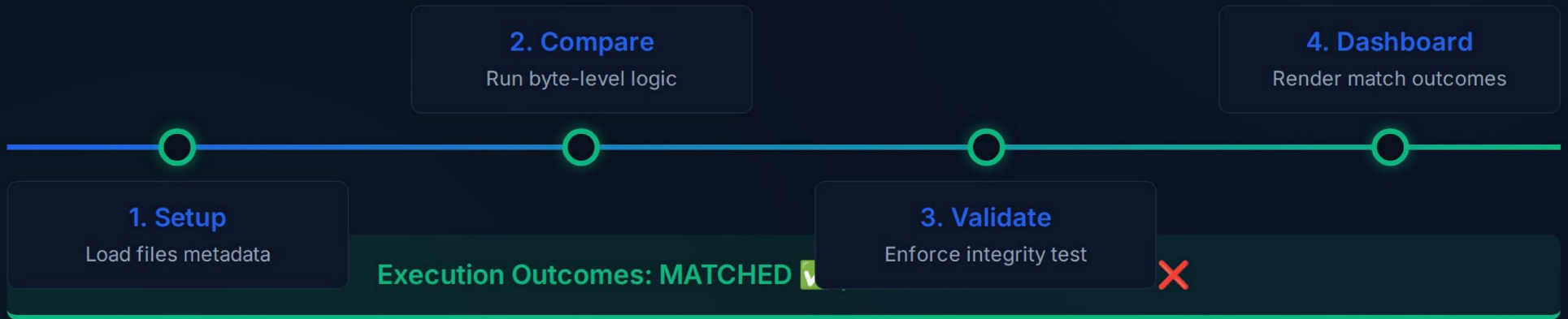
Task loop coordinator. Automates comparative threads sequentially inside system memory.



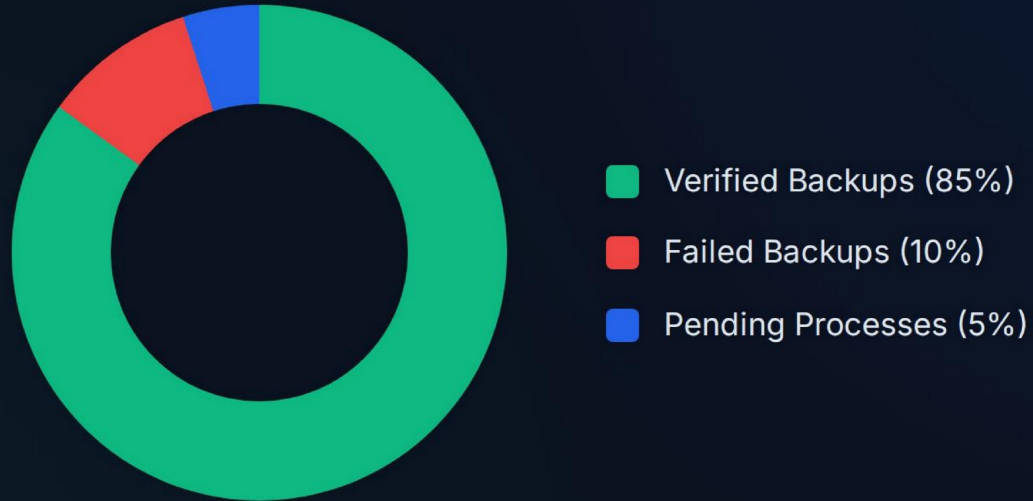
narrator.py

Reporting utility. Assembles clean, user-friendly markdown records of run outcomes.

• Verification Process



- **Monitoring Dashboard**



Real-time status analysis telemetry keeps critical server clusters visible with zero administration overhead.

• Future Scope

Technical Additions

- **SHA-256 Checksums:** Upgrade byte verification to cryptographic hashing.
- **Direct Cloud Connectors:** Direct validation support for S3/Azure blobs.
- **Scheduled Routines:** Native cron-like job execution planner.

Reporting Scope

- **Automated Mail Alerts:** Trigger e-mails automatically during mismatches.
- **Structured PDF Exporter:** Deliver PDF reports to technical risk auditors.
- **AI Anomaly Tracker:** Track trends to predict storage failures.

Moving toward enterprise-grade backup intelligence.

• Conclusion

Fully Automated Verification

Secures business backup procedures while removing the risk of human oversight.

Reduced Manual Overhead

Saves IT admins hours of daily manual validation checks.

Stronger Recovery Readiness

Guarantees disaster files are error-free and immediately bootable.

Sleek Visual Governance

Central dashboard aggregates operational indicators cleanly in one window.

Reliable backups require reliable verification.

Thank You — Questions & Answers Session